



**Michigan
Technological
University**

Progress Report 1:

Course Name: Clinical Decision Support and AI Modeling

Course Code: SAT 5141

Instructor: Guy Hembroff

Submitted by: Valeria Mudzindiko & Lusubilo Nyondo

Group No: 7

1. Introduction

Cervical cancer remains one of the leading causes of cancer-related deaths among women globally, despite being largely preventable through vaccination and early detection. Persistent infection with high-risk human papillomavirus (HPV) genotypes is the primary etiological factor (Dellino et al., 2024). However, disparities in access to screening and vaccination continue to result in high mortality rates, especially in low- and middle-income countries (Liu et al., 2021). Conventional screening techniques such as Pap smears, colposcopy, and biopsy remain effective but are often resource-intensive, subjective, and less accessible in developing regions.

The integration of Artificial Intelligence (AI) and Machine Learning (ML) has emerged as a transformative approach to enhance the precision, accessibility, and efficiency of cervical cancer diagnosis. Recent developments demonstrate that ML can outperform traditional diagnostic methods in predictive accuracy and speed. For instance, Roy et al. (2025) proposed an interpretable AI framework using stacking ensembles combined with expert knowledge, achieving 98% accuracy and improved transparency in prediction. Similarly, Al Mudawi and Alazeb (2022) used classical ML algorithms such as Random Forest, Gradient Boosting, and Support Vector Machines to achieve near-perfect accuracy in cervical cancer prediction using demographic and clinical data.

Deep learning models have also shown significant promise in image-based diagnostics. Sompawong et al. (2019) developed a Mask R-CNN model for automated Pap smear screening, achieving over 91% sensitivity and specificity in detecting abnormal cervical cells. Likewise, Liu et al. (2021) applied a ResNet-based computer-aided diagnosis (CAD) system to classify colposcopy images, demonstrating diagnostic performance comparable to expert colposcopists. These studies highlight how AI can reduce subjectivity, improve reproducibility, and alleviate the burden on specialists, particularly in low-resource healthcare systems.

Furthermore, Dellino et al. (2024) emphasized that AI can enhance not only image-based screening but also the stratification of HPV-positive patients by integrating multifactorial data including microbiota composition and immune status to develop personalized risk profiles. Complementarily, Rahimi et al. (2023) performed a systematic review of machine learning algorithms in cervical cancer survival prediction, noting that ensemble and deep learning models significantly improve prognostic accuracy compared to traditional statistical methods. These advancements point to a future where AI enables early detection, personalized treatment planning, and predictive analytics in oncology.

Building on these foundations, this study aims to develop a cervical cancer risk-to-timeline prediction model using machine learning techniques that estimate not only the likelihood but also the temporal progression from risk exposure to Cervical

Intraepithelial Neoplasia (CIN). By integrating interpretable AI models and clinical data, this project seeks to provide a reliable and transparent Clinical Decision Support System (CDSS) that enhances early intervention and preventive care for at-risk populations.

2. Methodology

The proposed study employs a machine learning-based approach to predict cervical cancer progression using demographic, behavioral, and clinical features. The goal is to develop an interpretable AI model that not only identifies women at risk but also estimates the likely time frame of disease progression.

2.1 Model Selection:

- Logistic Regression and Random Forest will serve as baseline classifiers for binary prediction (CIN=0 vs CIN=1).
- Gradient Boosting (XGBoost) will be explored to enhance performance and handle class imbalance.
- For time-based predictions, a Survival Analysis model such as Cox Proportional Hazards or Gradient Boosted Survival Trees will be implemented to estimate progression timelines.

2.2 Python Tools and Libraries:

- Data analysis and preprocessing: Pandas, NumPy, Scikit-learn
- Model building: Scikit-learn, XGBoost, Lifelines (for survival analysis)
- Visualization: Matplotlib, Seaborn

2.3 Model Evaluation:

- Classification metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC
- Survival model metrics: Concordance index (C-index) and calibration plots
- Cross-validation: 5-fold cross-validation will be used to ensure model generalizability.

2.4 Interpretability:

To ensure transparency in clinical decision-making, SHAP (Shapley Additive Explanations) and feature importance plots will be used to explain how each feature contributes to the prediction outcomes.

3. Dataset Description and Preprocessing

The study utilizes a publicly available dataset from the UCI Machine Learning Repository, containing 858 samples and 36 attributes related to cervical cancer risk factors. The [dataset](#) includes demographic, behavioral, and medical variables such as age, smoking, sexual partners, contraceptive use, and test results (Hinselmann, Schiller,

Cytology, Biopsy). Data preprocessing steps completed so far include handling missing values, removing duplicates, and assessing data balance.

3.1 Handling Missing Data

3.1.1 Coerce object to numeric value

The dataset contained entries with missing values, represented by “?”. First, these entries were converted from “?” to NaN to indicate the null values to ensure consistency and analytical accuracy.

3.1.2 Drop Attributes and compute values

The features were rated according to percentage of missing values. All Features (2 columns) with >90% of missing values were dropped and the other columns with <90% were imputed using median imputation techniques. This approach was chosen because the median is robust to outliers and provides a more representative measure of central tendency for skewed or non-normally distributed data compared to the mean.

3.2 Data Cleaning

Duplicate records were assessed to ensure data quality before model development using the duplicated() function. A total of 25 duplicate entries were identified and visualized in a bar graph to show the frequency of repetitions across the dataset. Since all duplicated rows contained identical values, one instance of each is retained while the rest were dropped. This process helps to maintain data accuracy and prevent bias in the predictive model

3.3 Data Balance Analysis

The balance assessment shows that the dataset is highly imbalanced, with only one positive case (CIN = 1) against a large number of negative cases (CIN = 0). This indicates that most women in the dataset did not have biopsy-confirmed cervical intraepithelial neoplasia (CIN). Such imbalance can significantly affect model performance, as most algorithms may learn to predict the majority class only, leading to poor sensitivity and inaccurate risk predictions. While SMOTE (Synthetic Minority Oversampling Technique) and other balancing strategies have been identified as the next step, they have not yet been applied in this phase of the project.

4. Future Work

The next phase of this project will focus on data validation, application of data balancing strategies, and dataset splitting to prepare the data for model development. Data validation will involve assessing completeness, consistency, and accuracy to ensure that all imputed values and variable formats align with analytical requirements.

Once validated, appropriate data balancing techniques, such as SMOTE or class weighting, will be applied to address the extreme class imbalance observed in the target variable and enhance model generalizability. Finally, the dataset will be split into training and testing subsets to support robust model training, cross-validation, and unbiased performance evaluation in subsequent stages.

Feature selection methods such as Recursive Feature Elimination (RFE) will be used to identify the most relevant predictors. Model evaluation will be conducted on a test set to assess accuracy and generalization performance. Finally, results will be visualized and interpreted to highlight key risk factors influencing cervical cancer progression and timelines.

References

- Al Mudawi, N. and Alazeb, A. (2022) 'A Model for Predicting Cervical Cancer Using Machine Learning Algorithms', *Sensors*, 22(11), p. 4132.
<https://doi.org/10.3390/s22114132>
- Dellino, M. et al. (2024) 'Artificial Intelligence in Cervical Cancer Screening: Opportunities and Challenges', *AI*, 5(4), pp. 2984–3000.
<https://doi.org/10.3390/ai5040144>
- Liu, L. et al. (2021) 'Computer-aided Diagnostic System Based on Deep Learning for Classifying Colposcopy Images', *Annals of Translational Medicine*, 9(13), p. 1045.
<https://doi.org/10.21037/atm-21-885>
- Rahimi, M., Akbari, A., Asadi, F. and Emami, H. (2023) 'Cervical Cancer Survival Prediction by Machine Learning Algorithms: A Systematic Review', *BMC Cancer*, 23, p. 341. <https://doi.org/10.1186/s12885-023-10808-3>
- Roy, P., Hasan, M., Islam, M. and Uddin, M. (2025) 'Interpretable Artificial Intelligence (AI) for Cervical Cancer Risk Analysis Leveraging Stacking Ensemble and Expert Knowledge', *Digital Health*, 11, pp. 1–31. <https://doi.org/10.1177/20552076251327945>

Sompawong, N. et al. (2019) ‘Automated Pap Smear Cervical Cancer Screening Using Deep Learning’, *Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 7044–7048.

<https://doi.org/10.1109/EMBC.2019.8856369>